Jagannath Narayanaswamy

Tel: 571-237-0592 | Email: jagannathn@gwmail.gwu.edu | LinkedIn | Github | Portfolio | Washington, DC

EDUCATION

George Washington University **2024 - 2026**

Master of Science in Data Analytics (SEAS Dean's Award) *Washington, DC*

PROFESSIONAL EXPERIENCE

Center for Community Resilience **Jan 2025 – Present**

Data Analyst *Washington, DC*

- Built React-PostgreSQL analytics platform analyzing economic data across 200+ census tracts, processing 500K+ demographic records with real-time filtering and interactive visualizations.
- Developed ETL pipelines integrating Census Bureau, FRED, and HUD APIs using Python (Pandas, NumPy), automating data refresh cycles and reducing manual processing from 40 hours to 8 hours weekly.
- Conducted time-series analysis and forecasting using ARIMA and LSTM models on 10+ years of historical data to project demographic and housing trends with 92% accuracy.
- Wrote SQL queries and stored procedures to aggregate multi-source datasets, creating dashboards that identified $2.3M in resource allocation opportunities and supported strategic planning decisions.

Fitzhugh Mullan Institute for Health Workforce Equity, GWU **Aug 2024 – Present**

Research Specialist III (Data Scientist) *Washington, DC*

- Developed interactive Medicaid Primary Care Workforce Tracker using React, JavaScript, D3.js to visualize provider distribution patterns across 3,000+ counties used by 100+ policymakers for quarterly workforce planning meetings.
- Engineered scalable ETL pipeline with Python to clean, transform, and validate Medicaid provider datasets, implementing data quality checks that reduced errors by 95% and eliminated 40 hours of manual validation.
- Performed regression analysis on workforce data (n=3227 counties), revealing statistically significant inverse correlation (R^2=0.78) between reimbursement rates and provider participation, validating findings through confidence interval analysis before presenting findings to policymakers.

TECHNICAL SKILLS

Programming Languages: SQL (window functions, CTEs, joins), Python, R, JavaScript, C
Data Visualization: Tableau, Power BI, D3.js, Matplotlib, Seaborn
Data Analysis: Pandas, NumPy, Excel (VLOOKUP, INDEX-MATCH, PivotTables, Power Query), Regression Analysis, Statistical Analysis, A/B Testing, Hypothesis Testing, Correlation Analysis
AI/ML: Supervised Learning (Random Forest, XGBoost, Logistic Regression), Time Series Forecasting, NLP, K-Means, Feature Engineering
Databases & ETL: PostgreSQL, MySQL, MongoDB, DuckDB, ETL Pipeline Development
Tools & Cloud: Git, AWS, Google Cloud, Docker, Streamlit, FastAPI, Jupyter, PySpark

PROJECTS

Support Ticket Intelligence & Automation Platform **GitHub**

- Performed exploratory analysis on 10,000+ support tickets using Python and SQL, applying NLP techniques (TF-IDF, text classification) to identify routing bottlenecks and building classification model (91% accuracy) that reduced manual categorization from 6 hours to 4.5 hours daily.
- Created interactive dashboard with 8 operational KPIs (resolution time, ticket volume trends, agent performance metrics) enabling leadership to optimize staffing allocation and reduce backlogs by 25%.

NYISO Market Analytics & Demand Forecasting **GitHub**

- Queried 2M+ hourly pricing records using SQL window functions and multi-table joins to analyze seasonal demand patterns, identifying $450K annual revenue opportunity through dynamic pricing during high-demand periods.
- Built time-series forecasting models (Random Forest R^2=0.91, LSTM RMSE=1.76) predicting 7-day electricity price movements with 92% accuracy and deployed anomaly detection flagging 47 congestion events for grid operators.

AI-Powered Clinical Triage System **Live | GitHub**

- Built enterprise healthcare platform using OpenAI GPT API with prompt engineering to analyze patient symptoms across 150+ conditions, achieving 99.2% diagnostic accuracy and reducing unnecessary ER visits by 40% for 10,000+ users.
- Developed HIPAA-compliant production system with FastAPI backend, React frontend, voice input capabilities, and emergency detection, delivering ¡2min symptom assessments with automated provider matching and telehealth integration.

Financial Risk Intelligence Platform **GitHub**

- Extracted and analyzed text data from 500+ corporate SEC 10-K filings using Python, identifying patterns in risk language (litigation mentions, uncertainty terms) and performing correlation analysis revealing companies with deteriorating tone experienced 14% average stock decline within 90 days.
- Developed automated quarterly reporting system tracking financial metric changes across portfolios, generating risk scoring dashboards that streamlined analyst workflow and improved investment screening efficiency by 40%.

CERTIFICATIONS & AWARDS

- GitHub Sponsored Hackathon Winner [Deer Valley Coding Foundation]
- Google Data Analytics Professional Certificate link
- Crash Course on Python link